## ITM University Gwalior

### School of Computing and Intelligent Systems

**Project-Based Learning (PBL)**

## Mini Project Report

# CAPTAIN IMPACT INDEX (CII)

### Analysis of Indian Men's ODI Captains — 2005 to 2025

| Project Detail | |
|---|---|
| Course | Data Analytics Using Python |
| Course Code | CSL0312 |
| Program | B.Tech. CSE (Artificial Intelligence & Machine Learning) |
| Semester | III |
| Academic Session | July–Dec 2026 |
| Domain | Sports Analytics |
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| Date of Submission | 16 October 2026 |

**Submitted as a Mini Project Report for Project-Based Learning (PBL)**

## 1. Introduction & Problem Statement

### 1.1 Introduction

**Cricket creates large quantities of structured information, including match dates, teams, venues, toss outcomes, competition labels and results. Data analytics can organize these records into a consistent dataset and reveal patterns that are difficult to compare by reading individual scorecards. This project studies Indian men's One Day International (ODI) matches from 2005 through 2025.**

**The project develops the Captain Impact Index (CII), a weighted summary of five dimensions: overall win performance, results in selected pressure competitions, overseas performance, outcomes in matches where the captain won the toss, and home performance. Component scores are expressed as percentages and combined using fixed weights.**

### 1.2 Real-World Problem

**Captaincy discussions often rely on isolated statistics or personal impressions. A captain may perform differently across home and overseas matches or in major competitions. This project tests a transparent, reproducible way to summarize several match-outcome dimensions. It is a learning model, not a complete measure of leadership.**

### 1.3 Problem Statement

**To collect, clean, and analyze Indian men's ODI match records from 2005–2025, assign captain and match-context labels, calculate five component scores and a weighted CII for five selected captains, and communicate the results using charts and an interactive dashboard.**

### 1.4 Motivation and Significance

- **Transforms public sports data into a structured analytical dataset.**
- **Demonstrates the effect of preprocessing choices on results.**
- **Combines multiple measurable dimensions in a transparent index.**
- **Provides practical experience with Python, Pandas, NumPy, Matplotlib, Plotly and Streamlit.**
- **Creates an interactive dashboard suitable for project demonstration.**

### 1.5 Objectives

- **Acquire ODI JSON records from Cricsheet.**
- **Filter India men's ODI matches dated 2005–2025 inclusive.**
- **Create consistent result, captain, location and pressure fields.**
- **Calculate five component scores and weighted CII.**
- **Validate row counts, match IDs and expected captain records.**
- **Build charts and an interactive dashboard.**

• Document assumptions, limitations and reproducibility steps.

### 1.6 Research Questions

• How do selected captains compare under the chosen CII formula?

• How does win percentage differ between home and overseas matches?

• What do results in selected pressure competitions show?

• What are the strengths and limitations of one weighted score?

### 1.7 Scope and Boundaries

The analysis covers India men's ODIs from 2005 through 2025. The five selected captains are Sourav Ganguly, Rahul Dravid, MS Dhoni, Virat Kohli and Rohit Sharma. It does not include every captain in the period and does not isolate captaincy as a cause of match outcomes. Team strength, opposition, player availability, venue, and tournament conditions may influence the results.

### 1.8 Workflow

• Download and extract the source archive.

• Parse match metadata from JSON files.

• Filter to India men's ODIs within the date range.

• Validate outcomes, location labels, captain mapping and competition labels.

• Calculate captain-level component percentages.

• Combine component scores using predefined weights.

• Create visualizations and a Streamlit dashboard.

• Document findings and limitations.

### 1.9 Expected Outcome

Deliverables include a cleaned match-level dataset, captain-level scorecard, supporting charts, interactive dashboard, and detailed report. A reviewer should be able to understand how the final score was constructed and what it can and cannot support.

## 2. Dataset Description & Source

### 2.1 Data Source

The project uses Cricsheet's public ODI JSON archive. Cricsheet documents a structured JSON format with match information such as dates, teams, venue, competition/event, gender, match type, toss and outcome. Source: https://cricsheet.org/downloads/odis_json.zip. Format documentation: https://cricsheet.org/format/json/.

The source provides match records; captain mapping and some derived labels were added by the project. The captain assignments should therefore be treated as project-created annotations, not as a field directly verified by the source.

### 2.2 Why Match-Level Data?

**CII uses match outcomes and match context rather than individual deliveries. The JSON source contains detailed ball-by-ball information, but the analytical table has one row per match with the fields needed for the stated questions. Match-level aggregation reduces complexity while preserving dates, participants, winner, toss, venue, event, captain and derived categories.**

## 2.3 Dataset Selection and Filters

| Stage | Count | Explanation |
|---|---|---|
| Initial India ODI dataset, mixed gender | 599 | Period-filtered India ODI records before gender filtering. |
| Women's records excluded | 125 | Removed to match men's ODI scope. |
| Final men's ODI dataset | 474 | Cleaned match-level dataset. |
| Five-captain subset | 422 | Matches attributed to the five selected captains. |

## 2.4 Dataset Fields

| Field | Meaning | Purpose |
|---|---|---|
| Match_ID | Unique match identifier | Uniqueness checks. |
| Date / Year | Match date and year | Period filter and time analysis. |
| India / Opponent | Participating teams | Match context. |
| Venue / City | Ground and city | Location classification. |
| Toss_Winner | Team winning toss | Toss-associated analysis. |
| Toss_Decision | Bat/field decision | Contextual field. |
| Match_Winner | Recorded match winner | Outcome construction. |
| India_Result | Win, Loss or Tie | Component score calculations. |
| Season / Match_Type | Competition and format labels | Filtering and grouping. |
| Location_Type | Home or Overseas | Location component. |
| Captain | Project captain mapping | Captain-level aggregation. |
| Event_Name | Competition name | Pressure-match classification. |
| Pressure_Match | Binary pressure label | Pressure score. |
| Win_Score | Overall decisive-result win percentage | CII component. |
| Pressure_Score | Pressure competition win percentage | CII component. |
| **Field** | **Meaning** | **Purpose** |
| Overseas_Score | Overseas win percentage | CII component. |
| Toss_Decision_Score | Wins among matches captain won toss | CII component. |
| Home_Performance_Score | Home win percentage | CII component. |
| CII | Weighted composite score | Final comparison. |

## 2.5 Result Distribution

| Result | Count | Share of 474 |
|---|---|---|
| Win | 288 | 60.76% |
| Loss | 160 | 33.76% |
| Tie | 26 | 5.49% |

| | | |
|---|---|---|
| Total | 474 | 100.00% |

**The current result construction labels matches without a recorded winner as Tie. This should be checked against source outcome details before publication because a missing winner can also represent a no-result outcome depending on source structure. The project's reported values are preserved here.**

### 2.6 Home and Overseas Distribution

| Location type | Count |
|---|---|
| Overseas | 356 |
| Home | 118 |
| Unknown | 0 |
| Total | 474 |

**An Indian-city list was used to classify known Indian venues as Home. Some records with missing city data were reviewed and manually assigned using venue context. Because manual review affects classification, a future version should store corrections and supporting notes in a separate mapping file.**

### 2.7 Captain-wise Match Coverage

| Captain | Matches in Master Dataset |
|---|---|
| Sourav Ganguly | 8 |
| Rahul Dravid | 57 |
| MS Dhoni | 223 |
| Virat Kohli | 87 |
| Rohit Sharma | 47 |

**Captain assignments were manually mapped because the source JSON did not provide a consistent dedicated captain field for this analysis. The selected five-captain subset contains 422 rows. The full mapping should be independently reviewed before treating the results as a formal sports evaluation.**

### 2.8 Reproducibility

**The working project produced a base dataset, a captain-enriched dataset, a pressure-enriched dataset, a final CII scorecard, and a master match-level CSV. Colab paths are environment-specific. For reproducibility, preserve the downloaded archive, extraction script, mapping files, notebooks, CSV outputs and dashboard script.**

## 3. Data Cleaning & Preprocessing

### 3.1 Purpose

**Preprocessing converts raw source records into consistent rows. Incorrect date filters, inconsistent team names, unreviewed missing values, or inaccurate captain assignments can affect every later score. Each transformation should be explained and validated.**

### 3.2 Reading JSON Files

The archive contains individual JSON files. A loop reads each file, extracts match metadata and appends one dictionary per match. Optional fields should be checked before access.

```
import json from pathlib import Path json_folder =
Path("/content/odis_json_extracted/odis_json")
match_rows = [] for file_path in json_folder.glob("*.json"):
with open(file_path, "r", encoding="utf-8") as f:
match = json.load(f)
# Extract date, teams, venue, toss, outcome and event. #
Append one match-level dictionary to match_rows.
```

This is a teaching pattern, not a complete JSON parser. The full extraction logic and source-field normalization should be retained in the project notebook.

### 3.3 Date Parsing and Filtering

```
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Year"] = df["Date"].dt.year
df = df[df["Year"].between(2005, 2025)].copy()
```

Parsing dates into a standard datetime type enables reliable filtering and year-wise grouping. Invalid dates should be inspected rather than silently accepted.

### 3.4 India and Men's ODI Filtering

Records were restricted to matches involving India, ODI format, male participants, and years 2005–2025. The initial mixed-gender India dataset contained 599 records; removing 125 women's matches produced 474 men's ODI records.

### 3.5 Result Construction

```
def india_result(row): if
row["Match_Winner"] == "India":
return "Win" elif
pd.isna(row["Match_Winner"]):
return "Tie"
else:
return "Loss" df["India_Result"] =
df.apply(india_result, axis=1)
```

The current implementation labels missing winner values as Tie. In source data, tie and no-result can be distinct outcomes, so a more robust version should inspect the outcome object and preserve separate labels. The current reported result counts are not silently altered in this report.

### 3.6 Duplicate Validation

```
assert len(master_df) == 474 assert
master_df["Match_ID"].nunique() == 474 assert
not master_df["Match_ID"].duplicated().any()
```

The master dataset contained 474 rows and 474 unique match IDs. This confirms uniqueness by ID, but it does not by itself validate all labels or calculations.

### 3.7 Home/Overseas Classification

Known Indian cities were classified as Home; matches outside that set were classified as Overseas after venue review. The final recorded totals are 118 Home and 356 Overseas. The project should keep a list of city aliases and manual overrides to make the rule repeatable.

```
indian_cities = {
# Add the reviewed Indian city names used in the project.
}
def classify_location(city, venue): if
pd.notna(city) and city in indian_cities:
return "Home"
# Apply documented venue overrides for ambiguous records.
return "Overseas"
```

### 3.8 Captain Mapping

The captain field was created through manual mapping, including stand-in captain assignments. The mapping covered all 474 records. For auditability, the recommended mapping table should include Match_ID, Captain, evidence, and a review flag.

```
captain_map = {
# "match_id": "captain name"
}
df["Captain"] = df["Match_ID"].map(captain_map)
```

### 3.9 Pressure Classification

Pressure matches were defined by event label: ICC Cricket World Cup, World Cup, ICC Champions Trophy, and Asia Cup. This produced 83 pressure matches and 391 non-pressure matches.

| Competition | Count |
|---|---|
| Asia Cup | 27 |
| ICC Cricket World Cup | 27 |
| ICC Champions Trophy | 21 |
| World Cup | 8 |
| Total | 83 |

```
pressure_events = {
"ICC Cricket World Cup", "World Cup",
"ICC Champions Trophy", "Asia Cup"
}
df["Pressure_Match"] =
df["Event_Name"].isin(pressure_events).astype(int)
```

This is an operational definition. It does not claim that every match in those competitions had equal pressure or that matches outside them were unimportant. Sourav Ganguly had no pressure matches under this definition, so Pressure Score is N/A rather than zero.

### 3.10 Final Master Dataset

The final master file, `CII_Master_Match_Level_Dataset.csv`, contains 474 rows and 24 columns. It combines source-derived fields with project-created labels and the captain-level component scores.

| Validation item | Recorded result |
|---|---|
| Rows | 474 |
| Columns | 24 |
| Unique Match_ID values | 474 |

| | |
|---|---|
| Duplicate Match_ID records | 0 |
| Unknown location records | 0 |
| Five-captain subset | 422 |

# 4. Exploratory Data Analysis (EDA)

## 4.1 Purpose of EDA

**EDA describes the dataset before interpretation. It helps identify imbalances, missing values, unusual labels, uneven sample sizes and potential data-quality issues. It is descriptive and should not be confused with causal inference.**

## 4.2 Dataset Size and Composition

**There are 474 men's India ODI matches in the cleaned dataset and 422 matches in the five-captain subset. The difference reflects records attributed to other captains. These totals should be repeated consistently in the dashboard, report and presentation.**

## 4.3 Overall Results

| Result | Count | Percentage |
|---|---|---|
| Win | 288 | 60.76% |
| Loss | 160 | 33.76% |
| Tie | 26 | 5.49% |

**These percentages describe all 474 match labels. Component scores use different denominators and therefore should not be substituted with these overall percentages.**

## 4.4 Captain Match Coverage

**MS Dhoni has the highest number of mapped matches in the full dataset. Captains with fewer matches have less stable percentages because a small number of outcomes can change the rate considerably. The CII ranking should therefore be read with sample sizes and career periods in mind.**

## 4.5 Location Distribution

**Overseas matches account for 356 records and Home matches account for 118. Differences between home and overseas percentages may reflect venue conditions, opposition, schedule and era, not only captaincy.**

## 4.6 Pressure Distribution

**The 83 pressure matches are drawn from four competition labels. This subset is smaller than the non-pressure group of 391 matches, and individual captain sample sizes vary. Percentages should be shown with match counts whenever possible.**

## 4.7 EDA Validation Checklist

- **Check that Year is between 2005 and 2025.**
- **Check India, gender and format filters.**
- **Check unique Match_ID count and duplicate count.**
- **Inspect result labels, including tie versus no-result.**
- **Review missing cities and manual venue overrides.**

• Review missing captain assignments.

• Confirm pressure labels match the event-name rule.

### 4.8 EDA Interpretation

EDA describes the composition of the available records. A larger match count is not itself a higher performance score, and a high percentage from a small sample may be less stable than the same percentage from a larger sample.

## 5. Data Visualization & Observations

### 5.1 Purpose

Charts support comparison and make patterns easier to communicate. Matplotlib was used for static analysis plots; Plotly was used for interactive dashboard charts. Interactive charts support hover inspection and help a reviewer explore individual values.

### 5.2 CII Comparison

The CII bar chart shows captain names on the horizontal axis and CII on the vertical axis, using a 0–100 scale. Sorting from highest to lowest makes the ranking clear. The chart should always be presented with the note that the index is project-defined.

### 5.3 Five Criteria Comparison

The grouped bar chart compares all five components. This is essential because two captains can have similar final scores but different component profiles. A single composite number can hide strengths and weaknesses, so the component chart should accompany the CII ranking.

### 5.4 Home vs Overseas

The location chart compares win percentages for Home and Overseas groups. The denominator for this component excludes ties. If a category has no decisive matches, it should be displayed as unavailable instead of being interpreted as a genuine 0% win rate.

### 5.5 Pressure Results

A pressure-result chart counts wins, losses and ties among the selected competitions. Counts complement percentages because a 75% win rate across four matches does not carry the same amount of evidence as a 75% rate across forty matches.

### 5.6 Year-Wise Match Coverage

The year-wise line chart displays the number of matches led by each captain by calendar year. It describes coverage and tenure, not performance. A chart of match counts should not be described as a trend in success.

### 5.7 Interactive Captain Filter

The dashboard's select box allows a user to inspect one captain's component values in a table. This feature is useful in a viva demonstration: first show the overall ranking, then select a captain, and explain the component profile.

### 5.8 Plotly Code Example

```
fig = px.bar( cii_summary.sort_values("CII",
ascending=False), x="Captain", y="CII",
text="CII", title="CII Score by Captain"
) fig.update_traces(texttemplate="%{text:.2f}",
textposition="outside") fig.update_layout(yaxis_range=[0, 100],
height=500) st.plotly_chart(fig, use_container_width=True)
```

### 5.9 Visual Quality

**Use descriptive titles, readable labels, consistent scales and clear legends. If text overlaps, adjust chart height, label orientation or font size. Colour should support meaning and should not be the only way a viewer can distinguish categories.**

### 5.10 Observations

**The recorded CII ranking places Rohit Sharma first, followed by Virat Kohli, MS Dhoni, Rahul Dravid and Sourav Ganguly. Component values vary, demonstrating why the report includes both the final score and its inputs.**

**The project uses Matplotlib for static analysis charts and Plotly for interactive visualizations in the Streamlit dashboard. The dashboard is built with Streamlit and provides interactive captain filtering and multiple Plotly charts.**

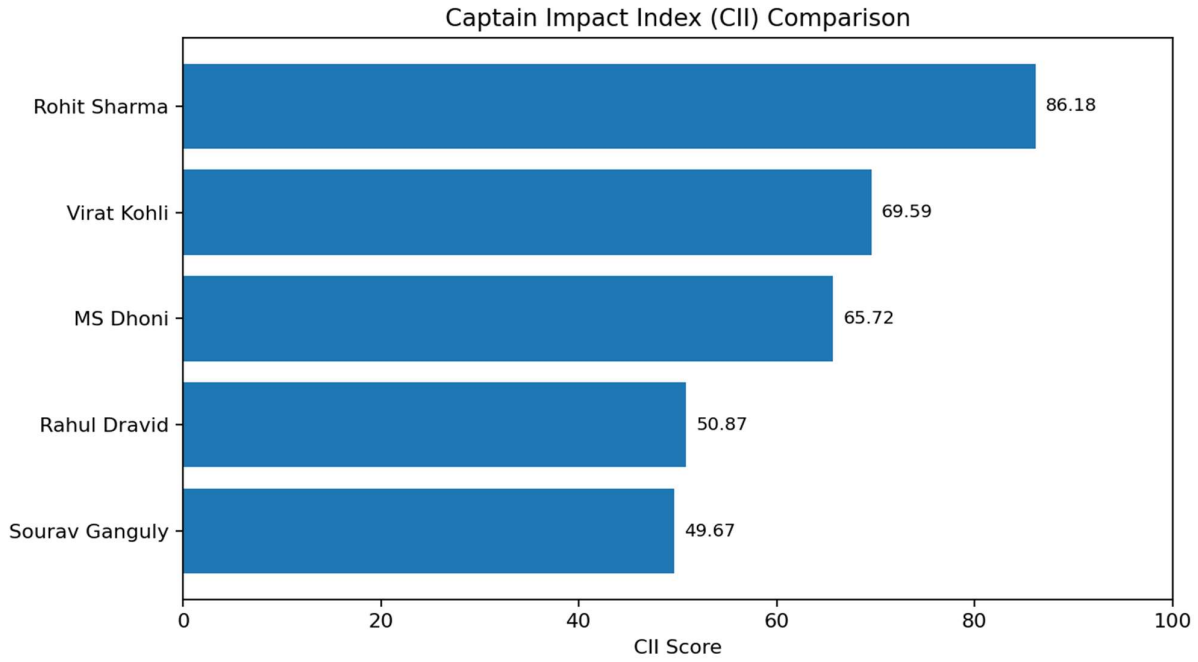



***Figure 1. Captain Impact Index (CII) comparison***

**Observation: Rohit Sharma has the highest CII (86.18), followed by Virat Kohli (69.59), MS Dhoni (65.72), Rahul Dravid (50.87), and Sourav Ganguly (49.67).**

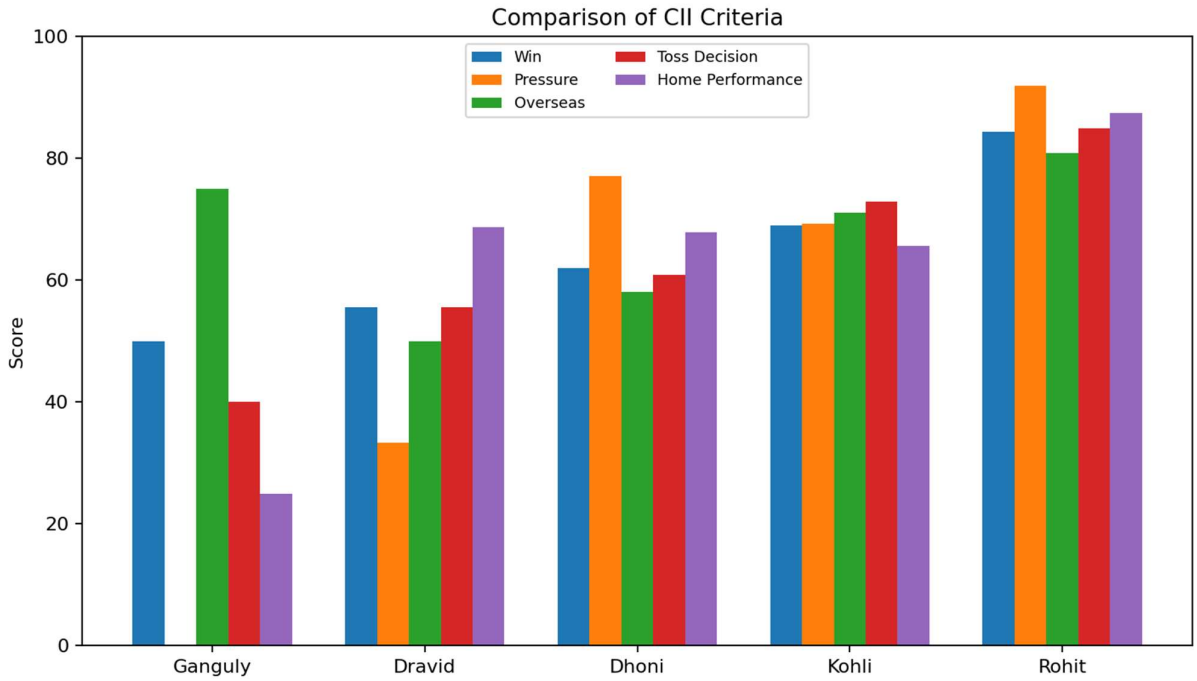



***Figure 2. Comparison of the five CII criteria***

**Observation: Rohit Sharma records the highest values across the five component scores in t**

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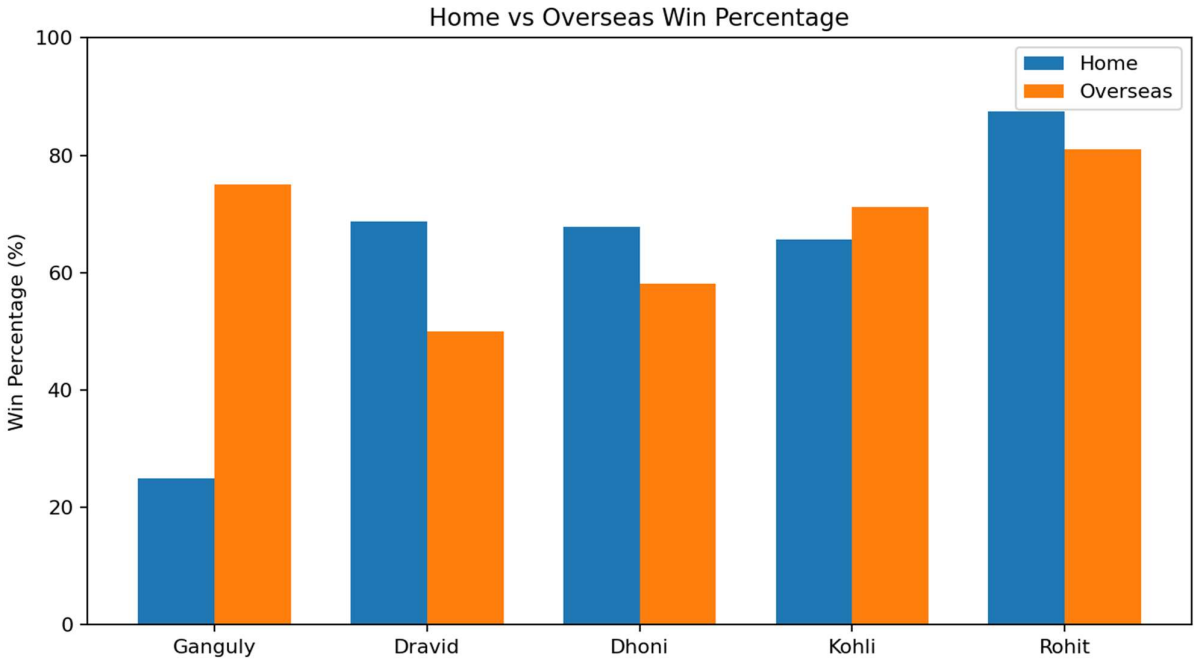



*Figure 3. Home vs Overseas win percentage*

**Observation: Rohit Sharma has the strongest home and overseas win percentages among the five selected captains. Sourav Ganguly has a relatively high overseas score but a lower home score in this dataset.**

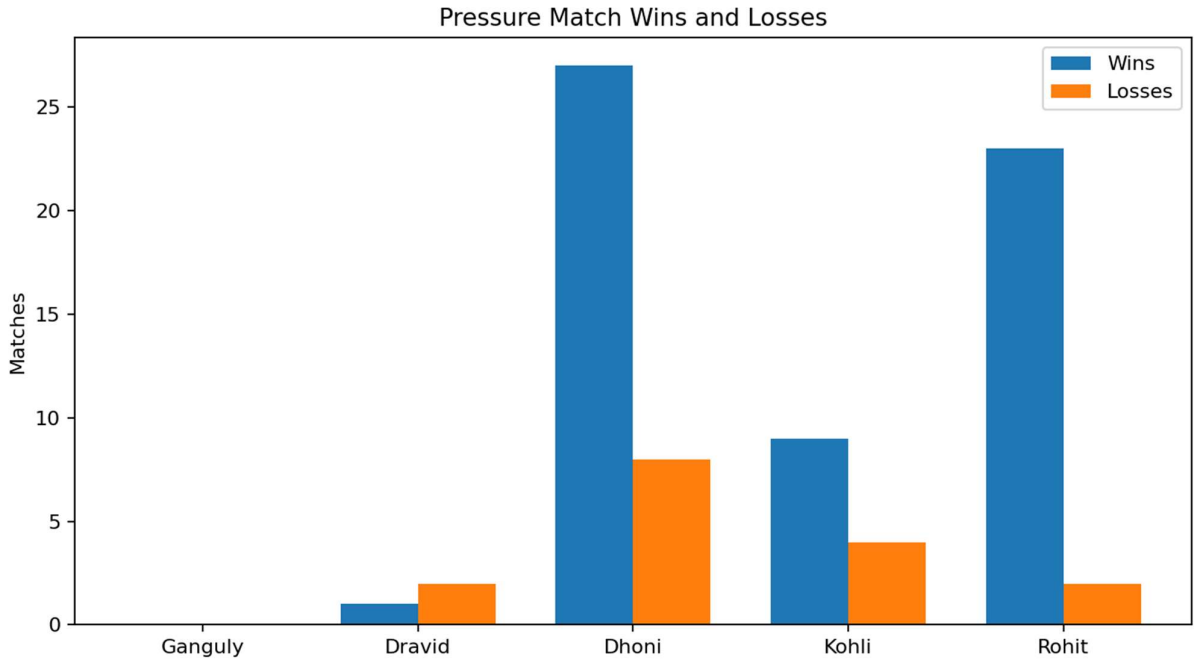



*Figure 4. Pressure-match wins and losses*

**Observation: Rohit Sharma records the highest values across the five component scores in this dataset except for no missing component; Ganguly's Pressure Score is unavailable because he has no pressure matches under the defined competition rule.**

## 6. Statistical Analysis

### 6.1 Purpose

**The analysis uses percentage-based components and descriptive statistics. These summaries describe the selected dataset; they do not prove that score differences are statistically significant or caused by captaincy.**

### 6.2 Win Percentage

**For Win Score, Pressure Score, Overseas Score and Home Performance Score, the implemented formula is:**

```
Score (%) = Wins / (Wins + Losses) × 100
```

**Ties are excluded from these four denominators. These are win percentages among decisive results, not percentages of all matches.**

### 6.3 Toss Decision Score

**The current Toss Decision Score is wins in matches where the captain won the toss divided by all matches where the captain won the toss, multiplied by 100. The current denominator includes ties. It measures an association with winning the toss, not a causal effect of the decision to bat or field.**

```
Toss Decision Score (%) = Wins after winning toss / Matches where captain won toss × 100
```

### 6.4 CII Formula and Weights

| Criterion | Weight | Meaning |
|---|---|---|
| Win Score | 25% | Overall decisive-result win percentage |

| Criterion | Weight | Meaning |
|---|---|---|
| Pressure Score | 25% | Win percentage in selected competitions |
| Overseas Score | 20% | Overseas decisive-result win percentage |
| Toss Decision Score | 15% | Win percentage when captain won toss |
| Home Performance Score | 15% | Home decisive-result win percentage |
| Total | 100% | All components available |

```
CII = 0.25*Win_Score + 0.25*Pressure_Score
+ 0.20*Overseas_Score + 0.15*Toss_Decision_Score
+ 0.15*Home_Performance_Score
```

### 6.5 Missing Pressure Score

**Ganguly had no qualifying pressure matches. His Pressure Score is N/A, not zero. The recorded CII calculation excludes the unavailable pressure weight and normalizes the remaining weights, whose total is 75%, to 100%. This should be disclosed because his score is calculated from a different available component set.**

### 6.6 Final Scorecard

| Captain | Win | Pressure | Overseas | Toss | Home | CII |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| Sourav Ganguly | 50.00 | N/A | 75.00 | 40.00 | 25.00 | 49.67 |
| Rahul Dravid | 55.56 | 33.33 | 50.00 | 55.56 | 68.75 | 50.87 |
| MS Dhoni | 62.02 | 77.14 | 58.06 | 60.91 | 67.86 | 65.72 |
| Virat Kohli | 69.05 | 69.23 | 71.15 | 72.97 | 65.62 | 69.59 |
| Rohit Sharma | 84.44 | 92.00 | 80.95 | 85.00 | 87.50 | 86.18 |

### 6.7 Worked Example

**Using the displayed rounded values, Rohit Sharma's CII is illustrated below. Small differences can arise if the original calculation used unrounded component values.**

```
CII = 0.25(84.44) + 0.25(92.00)
+ 0.20(80.95) + 0.15(85.00)
+ 0.15(87.50)
= 21.11 + 23.00 + 16.19 + 12.75 + 13.13
≈ 86.18
```

### 6.8 Descriptive Statistics

**Mean, median, minimum, maximum and range can summarize the five captain scores. These describe the selected observations only. Ganguly's missing pressure score must remain missing for pressure-specific summaries rather than being silently changed to zero.**

```
score_columns = [
"Win_Score", "Pressure_Score", "Overseas_Score",
"Toss_Decision_Score", "Home_Performance_Score", "CII"
]
print(cii_summary[score_columns].describe())
```

### 6.9 Limits of Inference

**Only five captain-level CII rows are compared. That is too small to support broad claims from correlation or regression using the aggregate scorecard alone. A stronger future model would define match-level outcomes and account for opponent, venue, tournament and team composition.**

### 6.10 Methodological Consistency Note

**The four win-rate components exclude ties, while the current Toss Decision Score includes ties in the denominator. This report preserves recorded values rather than silently recalculating them. A future revision should choose a consistent policy and regenerate the scorecard, charts and dashboard together.**

## 7. Key Insights & Interpretation

### 7.1 Highest CII in Dataset

**Based on the selected criteria and weighting methodology, Rohit Sharma obtained the highest CII score in this dataset, at 86.18. This statement refers to the recorded dataset and formula, not to an objective or universal judgement of captaincy.**

### 7.2 Virat Kohli's Profile

**Virat Kohli's CII is 69.59. The reported component values show performance across several included dimensions. Reviewing the components is more informative than describing only the aggregate score.**

### 7.3 MS Dhoni's Pressure Performance

MS Dhoni's Pressure Score is 77.14 and CII is 65.72. Under the project definition, this is a high win percentage among decisive results in selected competitions. The number should be read alongside sample size and tournament context.

### 7.4 Rahul Dravid's Profile

Rahul Dravid's CII is 50.87. His Home Performance Score is 68.75, while Overseas Score is 50.00 and Pressure Score is 33.33. This difference illustrates the value of inspecting component profiles.

### 7.5 Sourav Ganguly's Missing Component

Ganguly's CII is 49.67, but his Pressure Score is unavailable because there are no qualifying pressure matches in this sample. The CII normalizes remaining weights. His result is therefore not directly identical in component coverage to scores using all five criteria.

### 7.6 Ranking

| Rank | Captain | CII |
|---|---|---|
| 1 | Rohit Sharma | 86.18 |
| 2 | Virat Kohli | 69.59 |
| 3 | MS Dhoni | 65.72 |
| 4 | Rahul Dravid | 50.87 |
| 5 | Sourav Ganguly | 49.67 |

### 7.7 What the Index Can and Cannot Say

CII summarizes the chosen outcomes according to explicit rules. It cannot isolate leadership from teammate quality, opposition strength, injuries, scheduling, venue or era. Weights are design decisions, and a different set of weights may change the ranking.

### 7.8 Weight Sensitivity

A future sensitivity analysis could compare the current weighting with equal weights or alternative expert-selected weights. If the ranking changes, that would show that the result depends partly on model design. This report does not claim that such an analysis has already been completed.

### 7.9 Dashboard Value

The dashboard supports exploration from the overall ranking into criteria, location, pressure results and yearly coverage. This makes the analysis easier to explain and enables the reviewer to ask follow-up questions about particular components.

### 7.10 Presentation Cautions

- State that CII is project-defined, not official.
- State the 2005–2025 men's ODI scope.
- Explain that only five captains are ranked.

- **Disclose Ganguly's unavailable pressure score and normalized weights.**
- **Do not claim toss causality.**
- **Show match counts alongside percentages.**

## 8. Conclusion & Future Scope

### 8.1 Conclusion

**The CII project demonstrates an educational data analytics workflow from data acquisition through cleaning, feature creation, score calculation, visualization and dashboard deployment. The cleaned dataset contains 474 India men's ODI matches from 2005–2025; the selected five-captain subset contains 422 matches.**

**Under the current formula, Rohit Sharma ranks highest, followed by Virat Kohli, MS Dhoni, Rahul Dravid and Sourav Ganguly.**
**The report documents important caveats: captain mapping is manual, location classification includes reviewed assignments, Ganguly's pressure score is unavailable, and tie handling differs between components.**

### 8.2 Contributions

- **Cleaned match-level dataset.**
- **Documented pressure and location categories.**
- **Five-component weighted index.**
- **Captain scorecard and charts.**
- **Interactive Plotly/Streamlit dashboard.**
- **Reproducible files and validation recommendations.**

### 8.3 Limitations

- **Manual captain mapping requires audit.**
- **Some location labels required manual review.**
- **Pressure matches are competition-defined.**
- **Captains have unequal sample sizes and eras.**
- **Weights are project choices.**
- **Toss score is not a causal estimate.**
- **Tie denominator handling is inconsistent.**
- **Ganguly's score uses normalized available weights.**

### 8.4 Future Scope

- **Verify captain assignments independently.**
- **Store captain and venue corrections in versioned CSV files.**
- **Separate tie and no-result outcomes correctly.**

- **Standardize denominator rules.**
- **Adjust for opponent strength and neutral venues.**
- **Add sample-size warnings or uncertainty intervals.**
- **Compare alternative weights.**
- **Extend to other formats only after adapting definitions.**
- **Add downloadable filters and match-level drill-down.**
- **Automate refresh and validation.**

### 8.5 Final Recommendation

**The index is appropriate for an academic demonstration when its definitions and limits are clearly explained. Before using it as a formal evaluation tool, independently audit the captain mapping and resolve the denominator inconsistency.**

## 9. References


- **Cricsheet, ODI data downloads: https://cricsheet.org/downloads/odis_json.zip**
- **Cricsheet, JSON format: https://cricsheet.org/format/json/**
- **Pandas documentation: https://pandas.pydata.org/docs/**
- **NumPy documentation: https://numpy.org/doc/**
- **Matplotlib documentation: https://matplotlib.org/stable/**
- **Plotly Python documentation: https://plotly.com/python/**
- **Streamlit documentation: https://docs.streamlit.io/**
- **Python documentation: https://docs.python.org/3/**


## 10. AI Tool Usage Declaration

**AI tools were used during the project for coding assistance, debugging, documentation support, report organization, and explanation of data-analysis concepts. The team reviewed and adapted the generated material to the project's actual dataset and methodology.**

| AI Tool Used | Purpose | Extent of Use / Section Applied |
|---|---|---|
| ChatGPT | Coding guidance, debugging, explanation, report drafting | Python/Colab workflow, CII methodology explanation, dashboard guidance, report organization and documentation |

**☐ No AI tools were used in this project.**

**☑ AI tools were used, as detailed above.**

## 11. Project Links & Deliverables

**GitHub Repository Link:** _____

**Python Notebook (.ipynb) File Name:** _____

**PBL Report File Name:** _____

***Confirm that the GitHub repository includes the code, notebook, and a PBL report as required.***

☐ **GitHub repository includes code, notebook, *and a PBL report* and README**

## 12. Report Structure Checklist (Section 8 Compliance)

| S. No. | Report Section | Included |
|---|---|---|
| 1 | Title Page | ✓ |
| 2 | Introduction & Problem Statement | ✓ |
| 3 | Dataset Description & Source | ✓ |
| 4 | Data Cleaning & Preprocessing | ✓ |
| 5 | Exploratory Data Analysis (EDA) | ✓ |
| 6 | Data Visualization & Observations | ✓ |
| 7 | Statistical Analysis (Basic) | ✓ |
| 8 | Key Insights & Interpretation | ✓ |
| 9 | Conclusion & Future Scope | ✓ |
| 10 | References | ✓ |

## 13. Declaration by Student(s)

**We declare that this report presents the work carried out for the Captain Impact Index PBL mini project. We understand that CII is a project-defined analytical measure and that the data, code, results and interpretations must be checked before submission.**

| Name | Student ID | Signature | Date |
|---|---|---|---|
| R. Bharath | BETN1AI25203 | _____ | _____ |
| U. Naga Lokesh | BETN1AI25217 | _____ | _____ |
| M. Amar Deepika | BETN1AI25185 | _____ | _____ |

## 14. For Faculty Use Only — PBL Final Report Evaluation

**Evaluation Component B — PBL Final Report Submission | Mid-Term (PBL Based): 10 Marks**

| Evaluation Criteria | Max Marks | Marks Awarded |
|---|---|---|
| Data Cleaning & Preprocessing | 2 | |
| Exploratory Data Analysis (EDA) | 2 | |
| Visualization & Observations | 2 | |
| Statistical Analysis | 2 | |
| Insights, Interpretation & Conclusion | 2 | |
| Total | 10 | |

**Faculty Remarks:** _____

_____

**Faculty Mentor Signature:** _____

**Date:** _____